

SANJAY K

Curriculum Vitae

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Personal Information

I have successfully completed the Ph.D. viva voce in Mathematics at The Gandhigram Rural Institute (Deemed to be University), Gandhigram, under the supervision of Dr. P. Balasubramaniam, Senior Professor of Mathematics. My doctoral research, titled "Stabilization and Synchronization of Nonlinear Dynamical Systems using Hybrid Quantum Algorithm", focuses on advanced control-theoretic methodologies for complex nonlinear and uncertain dynamical systems. My research contributions lie at the interface of nonlinear dynamical systems, control theory, quantum-inspired computation, and data-driven intelligence. The thesis develops hybrid frameworks combining Lyapunov-based stability theory, linear matrix inequality optimization, quantum genetic algorithms, and physics-informed neural networks to address stabilization, synchronization, and filtering problems in interconnected and large-scale systems. Particular emphasis is placed on systems affected by time delays, memory effects, cyber-attacks, uncertainties, and fractional-order dynamics.

Date of Birth: 28-09-1997

Nationality: Indian

Education

- March 2022 - Feb 2026 **Doctor of Philosophy in Mathematics**, *The Gandhigram Rural Institute*, (Deemed to be University), Gandhigram, Dindigul, Tamil Nadu, India.
Title: Stabilization and Synchronization of Nonlinear Dynamical Systems using Hybrid Quantum Algorithm.
- 2018-2020 **Master of Science in Mathematics**, *Guru Nanak College (Autonomous)*, Affiliated to University of Madras, Velachery, Chennai-42, Tamil Nadu, India.
- 2015-2018 **Bachelor of Science**, (*Physics, Mathematics, Electronics*), University College of Science, Tumkur University, Tumkur, Karnataka, India.

Research Experience

- April 2023 - Feb 2026 **Project Associate**, in the SERB - CRG project entitled "**Quantum Neural Network-based Adaptive Controller for Nonlinear Dynamical Systems and Applications**", under the Principal Investigator *Dr. P. Balasubramaniam*, Senior Professor of Mathematics, The Gandhigram Rural Institute (Deemed to be University), Gandhigram, Dindigul, Tamil Nadu, India.
- Key Responsibilities
 - Integrate classical control theory with quantum-inspired computation.
 - Model and simulate nonlinear dynamical systems.
 - Contribute to research publications, technical documentation, and data-driven analysis.
 - Assist in mentoring research scholars and organizing project-related academic workshops or seminars.

Research Interests

- * Nonlinear Dynamical Systems
- * Quantum Computing
- * Control Systems
- * Image Processing

Publications

★ **Current status of publications: Journal (6) / Under-review (2), Citations-48, h-index-4**

- 1 **K. Sanjay** and P. Balasubramaniam, Synchronising Coronary Artery System via Physics-Informed Neural Feedback Control, **Biomedical Signal Processing and Control**, 114, 109311, April. 2026. DOI: 10.1016/j.bspc.2025.109311. (SCIE, Impact factor: 4.9, Quartile - Q1).
- 2 **K. Sanjay**, R. Vijay Aravind, and P. Balasubramaniam, Quantum Genetic Algorithm based Memory State Feedback Control for T-S Fuzzy System, **The European Physical Journal Special Topics**, 234 (8), pp. 1407-1414, Aug. 2024. DOI:10.1140/epjs/s11734-024-01293-1. (SCIE, Impact factor: 2.6, Quartile - Q2).
- 3 **K. Sanjay**, R. Vijay Aravind, and P. Balasubramaniam, Stabilization Analysis for Nonlinear Interconnected System with Memory based Coupled Sampled Data Control via Quantum-Inspired Genetic Algorithm, **Information Sciences**, 676, 120857, Aug. 2024. DOI: 10.1016/j.ins.2024.120857. (SCIE, Impact factor: 6.8, Quartile - Q1).
- 4 **K. Sanjay**, R. Vijay Aravind, and P. Balasubramaniam, Extended Dissipative Filter Design for Discrete-Time Interconnected Fuzzy Systems with Time-Varying Delays Subject to Cyber Attacks, **Applied Mathematics and Computation**, 453, 128071, Sep. 2023. DOI: 10.1016/j.amc.2023.128071. (SCIE, Impact factor: 3.4, Quartile - Q1).
- 5 **K. Sanjay** and P. Balasubramaniam, Controllability of Hilfer Type Fractional Evolution Neutral Integro-differential Inclusions with Non-instantaneous Impulses, **Evolution Equations and Control Theory**, 12(2), pp. 600-625, April 2023. DOI: 10.3934/eect.2022043. (SCIE, Impact factor: 1.2, Quartile - Q1).
- 6 N. Ramesh Babu, **K. Sanjay**, and P. Balasubramaniam, EED: Enhanced Edge Detection Algorithm via Generalized Integer and Fractional-Order Operators, **Circuits, Systems, and Signal Processing**, 41(10), pp. 5492-5534, April 2022. DOI: 10.1007/s00034-022-02028-0. (SCIE, Impact factor: 2.0, Quartile - Q2).

Patents and Intellectual Property

P. Balasubramaniam, N. Ramesh Babu, and **K. Sanjay**, "System and Method for Generating Enhanced Image from a Low Contrast Image," Indian Patent Office, Application No. 202341042226, Filed on 23 June 2023, Status: Reply Filed – Application in Amended Examination.

Awards and Honors

- 1 Recipient of **The Sydney R. Parker Best Paper Award**, for the paper entitled "*Enhanced Edge Detection Algorithm via Generalized Integer and Fractional-Order Operators*," presented by **Circuits, Systems and Signals Processing**, Birkhäuser (Springer), which was selected as the best paper published in 2022 in the area of signal processing.
- 2 Recipient of **The M.N.S. Swamy Best Paper Award**, for the paper entitled "*Enhanced Edge Detection Algorithm via Generalized Integer and Fractional-Order Operators*," presented by **Circuits, Systems and Signals Processing**, Birkhäuser (Springer), which was selected as the best paper published in 2021 or 2022 irrespective of the subject area.

Certificate Course

- 1 NPTEL online certification course in **Introduction to Quantum Computing: Quantum Algorithms and Qiskit** with a consolidated score of 61% (ELITE) conducted by the Indian Institute of Technology, Madras, during July-August 2025.
- 2 Completed the one-month course on **Quantum Computing**, during May 03rd–25th, 2025, organized by the Centre for Development of Advanced Computing, Hyderabad & Indian Institute of Technology, Roorkee, with the support of Ministry of Electronics and Information Technology, Government of India.

- 3 NPTEL online certification course in **Quantum Algorithms and Cryptography** with a consolidated score of 57% conducted by the Indian Institute of Technology, Madras, during January-April 2025.
- 4 Completion of **Scilab Training** organized at The Gandhigram Rural Institute by Manohar S with course material provided by the Spoken Tutorial Project, Indian Institute of Technology, Bombay. Dr. P. Balasubramaniam from The Gandhigram Rural Institute invigilated this examination.

Skills

- Languages Strong reading, writing and speaking competencies in English.
- Coding Python, C++, Scilab.
- Misc Academic research, teaching, critical thinking, $\text{\LaTeX}$ typesetting and publishing.

Conference Paper Presented

- 1 Presented a paper entitled "**Stabilization of T-S Fuzzy Systems: A Quantum Genetic Approach to Sampled-Data Controller Design**" at the International Workshop & Conference on Mathematical Techniques for Machine Learning and Quantum Computing, IWCMLQC-2024, during December 05th-07th, 2024, organized by the Department of Mathematics, Cochin University of Science and Technology, Kochi, Kerala, India.
- 2 Presented a paper entitled "**Quantum Genetic Algorithm based Memory State Feedback Control for the T-S Fuzzy System**" at the 3rd International Conference on Recent Trends in Applied Mathematics with Computing (ICRTAMC-2023), during December 21st-23rd, 2023, organized by the Department of Mathematics, Koneru Lakshmaiah Education Foundation (Deemed to be University), Vaddeswaram, Andhra Pradesh, India.
- 3 Presented a paper entitled "**Extended Dissipative Filter Design for Discrete-Time Interconnected Fuzzy Systems with Time-Varying Delays**" at the 9th International Conference on Discrete Mathematics and Mathematical Modelling in Digital Era (ICDMMMDE-2023), during March 23rd-25th, 2023, organized by the Department of Mathematics, The Gandhigram Rural Institute (Deemed to be University), Gandhigram, Dindigul, Tamil Nadu, India.
- 4 Presented a paper entitled "**The Need For Quantum-Resistant Cyber Security: A Review**" at the National Conference on Communication Technology & Network Security, on October 11th, 2019, organized by Bachelor of Computer Application, KLE Society's S.Nijalingappa College, Rajajinagar, Bengaluru, Karnataka, India.

Conference/Seminars/Workshops/Webinars/FDP Participated

- 1 Participated in the **International Workshop on Quantum Mathematics and Imaging Smart Manufacturing and Industry 4.0 (QMI-2025)**, during December 15th-19th, 2025, organized by the Department of Mathematics, School of Advanced Sciences, Vellore Institute of Technology, Chennai, Tamil Nadu, India.
- 2 Participated in the **Industry Workshop on Quantum Computing**, held on January 04th, 2025, organized by the Computer Society of India, Kolkata Chapter, India.
- 3 Participated in the **Science Academies' Lecture Workshop on Applications of Differential Equations in Engineering - 2024**, during March 04th-05th, 2024, organized by the Department of Mathematics, PSNA College of Engineering and Technology, Dindigul, Tamil Nadu, India.
- 4 Participated in the **Workshop on Algebra, Analysis and Number Theory (WAANT-2024)**, during January 25th-27th, 2024, organized by the Department of Mathematics, The Gandhigram Rural Institute (Deemed to be University), Gandhigram, Dindigul, Tamil Nadu, India.
- 5 Participated in the **IP Awareness/Training program**, under National Intellectual Property Awareness Mission, on August 09th, 2023, organized by Intellectual Property Office, India. (IP Trainer : Mr. J. E. Moshe Dayan.)
- 6 Participated in the **5-Day FDP on Advanced and Technical Computing with MATLAB**, on July 25th-29th, 2023, organized by the Department of Mathematics, School of Engineering, Presidency College, Bengaluru, India.

- 7 Participated in the five days **National Workshop on Algebra and Analysis (WAA-2023)**, in online mode held during May 22nd–26th, 2023, organized by the Department of Mathematics, National Institute of Technology Tiruchirappalli, Tamil Nadu, India.
- 8 Participated in the **IP Awareness/Training program**, under National Intellectual Property Awareness Mission, on May 12th, 2023, organized by Intellectual Property Office, India.
- 9 Participated in the **Workshop on Mathematical Aspects in Bio Sciences and Engineering (WMBSE-2022)**, on November 10th-11th, 2022, jointly organized by the Department of Mathematics & Department of Biomedical Engineering, SRMIST, Kattankulathur, Tamil Nadu, India.
- 10 Participated in the **International Webinar on Recent Trends in Pure and Applied Mathematics**, on July 26th-27th, 2022, organized by the Department of Mathematics, The Gandhigram Rural Institute, Gandhigram, Dindigul, Tamil Nadu, India.
- 11 Participated in the **Online Workshop on Improving Research Writing using Grammarly**, on April 27th, 2022, organized by Dr. G. Ramachandran Library in association with Bridge People Technology Solutions, The Gandhigram Rural Institute, Gandhigram, Dindigul, Tamil Nadu, India.
- 12 Participated in the virtual **National Conference on New Trends in Applied Mathematics**, on October 11th-12th, 2021, organized by the Department of Mathematics, University Institute of Engineering & Technology (U.I.E.T), Chhatrapati Shahu Ji Maharaj University Kanpur, India.
- 13 Participated in the **National Conference on Recent Trends in Electronics and Communication (RTEC-2018)**, on February 7th, 2018, organized by the Department of Electronics, University College of Science, Tumkur University, Tumkur, Karnataka, India.

Academic Achievements

Participated and obtained first class (19th rank), in the **District Level Mathematics Olympiad Exam** on Dec. 22nd, 2012, organized as a part of the **National Mathematics Year-2012**, conducted by Vijnana Bindu in collaboration with Vasavi Vidhya Peeta, Chickpet, Tumkur, Karnataka, India.

Extracurricular Activities

Participated in a **National Level Photography Competition** on World Environmental Day, organized by the Department of Visual Communication, Guru Nanak College, Velachery, Chennai, Tamil Nadu, India.
(Topic: Nature Under Lockdown)

Referees

1. **Dr. P. Balasubramaniam**
Senior Professor of Mathematics,
Dean, School of Sciences,
The Gandhigram Rural Institute (Deemed to be University),
Gandhigram - 624302, Dindigul, Tamil Nadu, India.
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